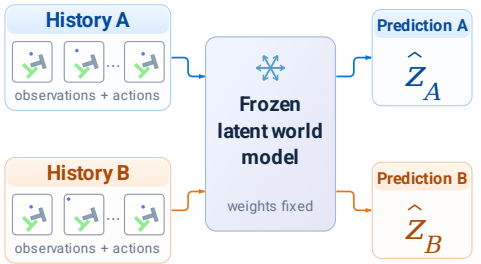
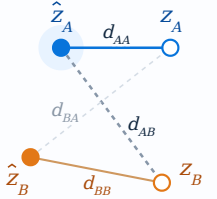
# B Paired evaluation protocol




## 1 Future discrimination

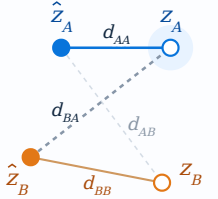
Hold the prediction fixed; compare the two true futures.




$d_{AA} < d_{AB}$ ; $d_{BB} < d_{BA}$

## 2 History benefit

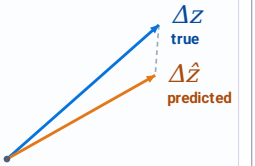
Hold the true future fixed; compare the two predictions.




$d_{AA} < d_{BA}$ ; $d_{BB} < d_{AB}$

## 3 Response fidelity

The predicted response should match the true response.




$$\text{NRE} = \frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}$$